



Math Foundations: Linear Algebra, Calculus & Statistics



Section 1: Linear Algebra

Q1. What is a vector and a matrix? How are they used in ML?

Answer:

A **vector** is a 1D array of numbers (direction + magnitude in space).

A **matrix** is a 2D array of numbers.

In ML:

- A **dataset** is a matrix X of shape $(n_samples \times n_features)$
 - **Weights** in a neural network layer are matrices W
 - A **data point** is a vector x of shape $(n_features,)$
 - **Dot product** $x^T w$ computes the weighted sum (linear model output)
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Q2. What is a dot product? What does it represent geometrically?

Answer:

Algebraic definition:

$$a \cdot b = \sum a_i b_i = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

Geometric definition:

$$a \cdot b = ||a|| \cdot ||b|| \cdot \cos(\theta)$$

Where θ = angle between vectors.

Meaning:

- If $a \cdot b = 0 \rightarrow$ vectors are **perpendicular** (orthogonal)
- If $a \cdot b > 0 \rightarrow$ angle $< 90^\circ$ (similar direction)

- If $\mathbf{a} \cdot \mathbf{b} < 0 \rightarrow \text{angle} > 90^\circ$ (opposite direction)

In ML: The dot product measures similarity. Used in attention mechanisms, SVM, and linear layers.

Q3. What is matrix multiplication? What are the rules?

Answer:

For $A (m \times n)$ and $B (n \times p)$:

$$C = A \times B \rightarrow \text{shape } (m \times p)$$

$$C_{ij} = \sum_k A_{ik} \cdot B_{kj}$$

Rule: Inner dimensions must match. A is $(m \times n)$ and B is $(n \times p)$.

Not commutative: $A \times B \neq B \times A$ (in general)

In neural networks: Each layer computes $\mathbf{z} = \mathbf{XW} + \mathbf{b}$ using matrix multiplication.

Q4. What are eigenvalues and eigenvectors?

Answer:

For a square matrix A , vector $\mathbf{v}$ is an eigenvector with eigenvalue λ if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

The matrix transforms $\mathbf{v}$ by only **scaling** it (not rotating).

Finding eigenvalues:

$$\det(A - \lambda I) = 0$$

In ML:

- **PCA** finds eigenvectors of the covariance matrix – they are the principal components
 - Larger eigenvalue $\rightarrow$ more variance explained by that component
 - **SVD** decomposes matrices for dimensionality reduction, recommendation systems
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Q5. What is a covariance matrix?

Answer:

The covariance matrix Σ captures how features **co-vary** with each other.

For dataset X with n features:

$$\Sigma_{ij} = \text{Cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)]$$

- **Diagonal entries** = variance of each feature
- **Off-diagonal entries** = covariance between feature pairs

Shape: ($n_{\text{features}} \times n_{\text{features}}$)

In PCA: We compute the covariance matrix, then find its eigenvectors to determine principal components.

Q6. What is the L1 and L2 norm? When do you use each?

Answer:

L1 Norm (Manhattan):

$$||x||_1 = \sum |x_i|$$

L2 Norm (Euclidean):

$$||x||_2 = \sqrt{\sum x_i^2}$$

In ML:

- **L2 norm** = standard Euclidean distance; used in KNN, SVM margin, gradient magnitude
 - **L1 norm** = produces sparse solutions; used in Lasso regularization
 - **L2 norm** in loss $\rightarrow$ Ridge regression; **L1 norm** in loss $\rightarrow$ Lasso regression
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Q7. What is a singular value decomposition (SVD)?

Answer:

SVD decomposes any matrix $A (m \times n)$ into:

$$A = U \cdot \Sigma \cdot V^T$$

Where:

- **$U (m \times m)$** – left singular vectors (orthogonal)
- **$\Sigma (m \times n)$** – diagonal matrix of singular values (non-negative, sorted descending)
- **$V^T (n \times n)$** – right singular vectors (orthogonal)

Applications in ML:

- **Dimensionality reduction** (Truncated SVD / LSA for NLP)
 - **Collaborative filtering** (recommendation systems)
 - **PCA** is a special case of SVD on centered data
 - **Image compression**
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Q8. What is the transpose of a matrix? What are its properties?

Answer:

Transpose of A ($m \times n$) is A^T ($n \times m$), obtained by swapping rows and columns.

Properties:

$$(A^T)^T = A$$

$$(AB)^T = B^T A^T$$

$$(A + B)^T = A^T + B^T$$

Symmetric matrix: $A = A^T$ (covariance matrices are symmetric)

Q9. What is an orthogonal matrix?

Answer:

A matrix Q is orthogonal if:

$$Q^T Q = Q Q^T = I \quad \rightarrow \quad Q^T = Q^{-1}$$

All columns are **unit vectors** and **mutually orthogonal**.

Importance: Orthogonal transformations preserve distance and angle — used in PCA (principal components are orthogonal), QR decomposition, and rotation matrices.

Q10. What is the rank of a matrix?

Answer:

The rank of matrix A is the number of **linearly independent** rows (or columns).

- **Full rank:** $\text{rank} = \min(m, n) \rightarrow$ matrix is invertible (if square)
- **Rank deficient:** $\text{rank} < \min(m, n) \rightarrow$ multicollinearity exists in data

In ML: If the feature matrix X is rank-deficient, the normal equations $(X^T X)^{-1} X^T y$ used in linear regression are not solvable. Ridge regression adds λI to make it invertible.

Section 2: Calculus for ML

Q11. What is a derivative? What is the chain rule?

Answer:

Derivative measures the rate of change of a function with respect to a variable:

$$f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$$

Chain rule: For composite function $f(g(x))$:

$$d/dx [f(g(x))] = f'(g(x)) \cdot g'(x)$$

In backpropagation:

$$\partial L / \partial w = (\partial L / \partial a) \cdot (\partial a / \partial z) \cdot (\partial z / \partial w)$$

Each term is computed layer by layer using the chain rule.

Q12. What is a partial derivative and a gradient?

Answer:

Partial derivative: Derivative with respect to one variable, holding others constant:

$$\partial f / \partial x_1 \quad (\text{holding } x_2, x_3, \dots \text{ constant})$$

Gradient: Vector of all partial derivatives:

$$\nabla f(x) = [\partial f / \partial x_1, \partial f / \partial x_2, \dots, \partial f / \partial x_n]$$

The gradient points in the direction of **steepest increase** of f . Gradient descent moves in the **negative gradient direction** to minimize f .

Q13. What is a local minimum vs global minimum?

Answer:

- **Global minimum:** The lowest point of the entire loss landscape
- **Local minimum:** A point where the loss is lower than all nearby points, but not globally the lowest
- **Saddle point:** Gradient = 0, but not a minimum in all directions

In practice: Deep learning models have high-dimensional loss landscapes. Most local minima are near the global minimum in quality. Momentum-based optimizers (Adam) help escape saddle points.

Q14. Derive the gradient of MSE loss.

Answer:

MSE loss:

$$L = (1/n) \cdot \sum (y_i - \hat{y}_i)^2 \quad \text{where } \hat{y}_i = w^T x_i$$

Gradient with respect to w:

$$\partial L / \partial w = (2/n) \cdot \sum (\hat{y}_i - y_i) \cdot x_i$$

In matrix form:

$$\partial L / \partial w = (2/n) \cdot X^T (Xw - y)$$

Setting this to zero gives the **Normal Equations:**

$$w^* = (X^T X)^{-1} X^T y$$

Q15. What is the Jacobian matrix?

Answer:

The Jacobian generalizes the gradient to vector-valued functions. For $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$:

$$J = [\partial f_i / \partial x_j] \quad (m \times n \text{ matrix})$$

In DL: Used in computing gradients through layers with vector outputs. The Jacobian of a softmax output with respect to its input is important for backprop through the softmax layer.

Q16. What is a convex function? Why does it matter?

Answer:

A function is **convex** if the line segment between any two points on the curve lies above or on the curve:

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) \quad \text{for } \lambda \in [0, 1]$$

Properties:

- Any local minimum is also the **global minimum**
- Gradient descent is guaranteed to converge

In ML:

- MSE loss for linear regression → convex (one global minimum)
- Loss for neural networks → **non-convex** (many local minima, saddle points)
- SVM dual problem → convex (guaranteed optimal solution)



Section 3: Statistics & Probability

Q17. What is the difference between probability and likelihood?

Answer:

	Probability	Likelihood
Fixed	Model parameters	Observed data
Varies	Data outcomes	Parameter values
Question	What data is likely given parameters?	What parameters best explain observed data?
Notation	$P(\text{data} \text{parameters})$	$L(\text{parameters} \text{data})$

Maximum Likelihood Estimation (MLE): Find parameters that maximize $L(\theta | \text{data}) = P(\text{data} | \theta)$.

Q18. What is Bayes' Theorem?

Answer:

$$P(A|B) = P(B|A) \cdot P(A) / P(B)$$

In ML terms:

$$P(\theta | \text{data}) = P(\text{data} | \theta) \cdot P(\theta) / P(\text{data})$$

$$\text{Posterior} = \text{Likelihood} \times \text{Prior} / \text{Evidence}$$

MAP (Maximum A Posteriori): Maximize $P(\theta | \text{data})$ – includes prior belief about parameters. L2 regularization is equivalent to MAP with a Gaussian prior.

Q19. What is a probability distribution? Name key ones.

Answer:

A probability distribution describes how probabilities are spread over possible outcomes.

Distribution	Type	PDF/PMF	Use in ML
Normal (Gaussian)	Continuous	$f(x) = (1/\sigma\sqrt{2\pi}) \cdot e^{\{-(x-\mu)^2/2\sigma^2\}}$	Feature normalization, noise assumption
Bernoulli	Discrete	$P(x) = p^x(1-p)^{1-x}$	Binary classification output
Binomial	Discrete	$P(k) = C(n,k) \cdot p^k \cdot (1-p)^{n-k}$	Count successes
Multinomial	Discrete	Generalization of Binomial	Multi-class classification
Exponential	Continuous	$f(x) = \lambda e^{\{-\lambda x\}}$	Time between events
Poisson	Discrete	$P(k) = (\lambda^k e^{\{-\lambda\}})/k!$	Count events in interval

Q20. What is hypothesis testing? Explain p-value.

Answer:

Hypothesis testing is a statistical method to make decisions about population parameters based on sample data.

Steps:

1. State **H₀**(null hypothesis) and **H₁**(alternative)
2. Choose significance level **α**(typically 0.05)
3. Compute test statistic
4. Compute **p-value**
5. If $p\text{-value} < \alpha \rightarrow \text{reject } H_0$

p-value: Probability of observing a result as extreme as the test statistic, **assuming H₀ is true**.

- Small p-value → data is unlikely under H₀ → evidence against H₀
- p-value is NOT the probability that H₀ is true

Q21. What is ANOVA?

Answer:

ANOVA (Analysis of Variance) tests whether means of **3+ groups** differ significantly.

One-way ANOVA:

- $H_0: \mu_1 = \mu_2 = \dots = \mu_k$ (all group means equal)
- H_1 : At least one group mean differs

F-statistic:

$$F = (\text{Between-group variance}) / (\text{Within-group variance})$$
$$= MSB / MSW$$

Large $F \rightarrow$ means differ more than expected by chance.

In your research: You used **ANOVA F** for feature selection – ANOVA F-score measures how much each feature's distribution differs across sentiment classes. High F-score $\rightarrow$ feature is discriminative $\rightarrow$ select it.

Q22. What are Type I and Type II errors?

Answer:

	Predicted Positive	Predicted Negative
Actually Positive	✓ True Positive	✗ Type II Error (False Negative)
Actually Negative	✗ Type I Error (False Positive)	✓ True Negative

- **Type I Error (α):** Rejecting H_0 when it's actually true (False Positive). Controlled by significance level α .
 - **Type II Error (β):** Failing to reject H_0 when H_1 is true (False Negative).
 - **Power = $1 - \beta$:** Probability of correctly detecting a real effect.
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Q23. What is the Central Limit Theorem?

Answer:

The CLT states that the **sampling distribution of the sample mean** approaches a **normal distribution** as sample size $n \rightarrow \infty$, regardless of the original population distribution.

$$\bar{X} \sim N(\mu, \sigma^2/n) \quad \text{for large } n$$

Why it matters in ML:

- Justifies using Gaussian assumptions in many statistical tests
- Underpins confidence intervals

- Explains why averaging predictions (ensembles) works well
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Q24. What is correlation vs causation?

Answer:

- **Correlation:** A statistical relationship between two variables (they tend to move together). Measured by Pearson's $r \in [-1, 1]$.
- **Causation:** One variable **directly causes** the other.

Pearson correlation:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum (x_i - \bar{x})^2] \cdot [\sum (y_i - \bar{y})^2]}}$$

"Correlation does not imply causation" – two variables may be correlated due to a **confounding variable**.

In ML: High correlation between features → multicollinearity issue. Between feature and target → useful predictor, but not necessarily causal.

Q25. What is the difference between mean, median, and mode?

Answer:

Measure	Formula	Best For
Mean	$\sum x_i / n$	Symmetric distributions, no outliers
Median	Middle value when sorted	Skewed distributions, robust to outliers
Mode	Most frequent value	Categorical data

In ML:

- Use **median imputation** for missing values when outliers are present
 - Use **mean imputation** for normally distributed data
 - **Standard deviation** $\sigma = \sqrt{(\sum (x_i - \mu)^2 / n)}$ measures spread
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Q26. What is the normal distribution and why is it important?

Answer:

The normal (Gaussian) distribution has the bell-curve shape:

$$f(x) = \frac{1}{(\sigma\sqrt{2\pi})} \cdot \exp\left(-\frac{(x-\mu)^2}{(2\sigma^2)}\right)$$

68-95-99.7 rule:

- 68% of data lies within 1σ of mean
- 95% within 2σ
- 99.7% within 3σ

Why important in ML:

- Many real-world features are approximately normal
 - Assumed in Linear Discriminant Analysis, Naive Bayes (Gaussian)
 - Residuals in linear regression should be normally distributed
 - Weight initialization in neural networks uses normal distribution (Xavier/He)
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Q27. What is Bayesian inference?

Answer:

Bayesian inference updates beliefs about parameters as new data arrives:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

$$P(\theta|X) \propto P(X|\theta) \cdot P(\theta)$$

Example – Naive Bayes classifier:

$$P(y|x_1, \dots, x_n) \propto P(y) \cdot \prod P(x_i|y)$$

"Naive" assumption: features are conditionally independent given the class.

Bayesian vs Frequentist:

- **Frequentist:** Parameters are fixed; probability is long-run frequency
 - **Bayesian:** Parameters are random variables; probability is degree of belief; prior knowledge incorporated
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Q28. What is the Chi-squared (χ^2) test?

Answer:

Chi-squared test checks whether observed frequencies differ significantly from expected frequencies.

Formula:

$$\chi^2 = \sum (O_i - E_i)^2 / E_i$$

Where O = observed frequency, E = expected frequency.

Uses:

1. **Goodness of fit test** – does data follow expected distribution?
 2. **Test of independence** – are two categorical variables independent?
 3. **Feature selection** (your research!) – χ^2 score measures dependence between feature and target class label. High $\chi^2 \rightarrow$ feature is informative $\rightarrow$ select it.
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